

IoT Sentinel - Security Assessment Report

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1. Executive Summary

Total Records Analyzed:	5000	Threats Detected:	1999
Benign Traffic:	2999	Critical Threat Events:	1281
Analysis Model:	XGBoost_Classifier_Core	Detection Accuracy:	99.90%

2. Critical Security Findings

- Intrusion activity detected: 1999 of 5000 records (40.0%) classified as malicious.
- Primary attack vector: DDoS was the most frequent incident category.
- Immediate action required: 1281 security events classified as CRITICAL severity.
- Model verification: Threat evaluation completed with a 99.9% test accuracy rating using the 'XGBoost' classifier.

3. Threat Profile Breakdown

Attack Category	Record Count	Percentage
DDoS	728	14.56%
Benign	151	3.02%
Mirai	516	10.32%
Reconnaissance	407	8.14%
Brute Force	349	6.98%

4. Recommended Mitigation Strategies

For DDoS Traffic:

- Isolate affected endpoint to prevent outbound amplification
- Rate-limit suspicious traffic incoming at network border
- Review source IP patterns and coordinate with upstream ISP

For Mirai Traffic:

- Isolate infected IoT node immediately (disconnect from network)

- Inspect active sessions for telnet/SSH brute force activity
- Reboot device to clear memory resident malware

For Reconnaissance Traffic:

- Investigate scanning source IP and log connection attempts
- Temporarily block scanner IP at host or network firewall
- Restrict unnecessary exposed services and scan targets

For Brute Force Traffic:

- Enforce strict rate-limiting on authentication endpoints
- Block attacking IP address temporarily at firewall
- Review accounts with high authentication failures

For Benign Traffic:

- No security action required.
- System remains under continuous monitoring.